Point-source solutions for fractional porous medium diffusion on complete manifolds

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Abstract

We construct solutions of the spectral fractional porous medium equation starting from a point mass on a complete noncompact Riemannian manifold. The assumptions are a lower Ricci curvature bound and a uniform Euclidean Faber–Krahn inequality. Every exponent $m > 1$, fractional order $0 < s < 1$, prescribed source point and positive mass is allowed. The solution is strongly continuous in L^1 at positive times, locally bounded, and has the prescribed initial trace against every bounded continuous function. The construction combines the known smoothing estimates for integrable data with averages of the linear heat semigroup. These averages have small fractional Laplacian uniformly in their spatial cutoff, which controls mass at infinity while passing to a point-source limit.

1 Introduction and main result

The porous medium equation with fractional diffusion describes slow nonlocal spreading of a concentration. A point-source solution starts with all the material concentrated at one location. On a manifold the geometry affects the diffusion, and the existence question can be asked without a radial coordinate or a spatial scaling symmetry.

For the Euclidean fractional porous medium equation, finite nonnegative measure data are covered by Grillo, Muratori and Punzo [3, Theorem 3.2]. Berchio, Bonforte, Grillo and Muratori [1] developed a theory on complete noncompact manifolds with a Ricci lower bound and a uniform Faber–Krahn inequality. Their construction gives weak dual solutions for integrable functions and for a larger weighted class, together with smoothing estimates. Existence for a Dirac initial datum is listed among the questions in [1, Section 7] and in Grillo's 2025 workshop report [4]. Here we prove existence in the distributional class stated below.

Throughout, (M, g) is a smooth complete, connected, noncompact Riemannian manifold of dimension $N \geq 2$, with Riemannian volume μ. Assume that

$$\mathrm{Ric} \geq -(N-1)k \quad \text{for some } k \geq 0, \qquad \lambda_1(D) \geq c\,\mathrm{Vol}(D)^{-2/N} \tag{1}$$

for some $c > 0$ and every relatively compact smooth domain $D \subset M$. Here $\lambda_1(D)$ is the first Dirichlet eigenvalue of $-\Delta_M$. Let L be the nonnegative Friedrichs realization of $-\Delta_M$ on $L^2(M)$ and let $B = L^s$, defined by the spectral calculus, where $0 < s < 1$. We consider

$$\partial_t u + B(u^m) = 0, \qquad m > 1. \tag{2}$$

The spaces $L^p(M)$ use μ, and $C_b(M)$ denotes bounded continuous functions. The distributional equation means

$$\int_0^\infty \int_M u\,\partial_t\phi\,d\mu\,dt = \int_0^\infty \int_M u^m B\phi\,d\mu\,dt, \qquad \phi \in C_c^\infty((0,\infty)\times M), \tag{3}$$

with an absolutely integrable right side.

Theorem 1.1. *Assume* (1). *For every* $m > 1$, $0 < s < 1$, $o \in M$ *and* $A > 0$, *there is a nonnegative solution of* (3) *such that*

$$u \in C((0,\infty); L^1(M)) \cap L^\infty_{\mathrm{loc}}((0,\infty)\times M), \qquad u^m \in L^1_{\mathrm{loc}}((0,\infty)\times M), \tag{4}$$

and, for every $T > 0$,

$$\sup_{0<t<T} \|u(t)\|_1 = A > 0. \tag{5}$$

Its initial trace is $A\delta_o$ *in the sense that*

$$\lim_{t\downarrow 0} \int_M u(t)\psi \, d\mu = A\psi(o) \qquad \text{for every } \psi \in C_b(M). \tag{6}$$

The assertion concerns existence on each manifold satisfying (1). Uniqueness and a common large-time profile are separate questions. The proof uses the integrable-data semigroup and smoothing estimate of [1]. The additional step is to pass from concentrating initial functions to a curve in the global space $L^1(M)$.

To do this, we first obtain local compactness at positive times from an energy estimate for u^m. Averages of the linear heat semigroup then give tightness over each finite time interval. At suitable positive times these two facts yield convergence in $L^1(M)$, and the contraction of the same nonlinear semigroup propagates that convergence to all later times. Finally, the smoothing estimate makes the nonlinear flux integrable down to time zero, which identifies the trace. The argument does not use a Green-potential uniqueness theorem.

2 Bounded-data solutions and their estimates

Write $P_r = e^{-rL}$ for the linear heat semigroup. Under (1), its symmetric kernel $p_r(x,y)$ satisfies

$$0 \le p_r(x,y) \le Cr^{-N/2} \exp\left(-\frac{d(x,y)^2}{Cr}\right), \qquad r > 0, \tag{7}$$

by [1, Assumption 2.1 and (2.3)]. Constants denoted by C may change from line to line. The Ricci bound also implies stochastic completeness, so $P_r 1 = 1$; see [2, Theorem 15.4(a)]. For example, the radial curvature condition there holds with $q(r) = \sqrt{k}+1+r$, and $\int^\infty dr/q(r) = \infty$.

The Faber–Krahn assumption in [1] is stated for arbitrary relatively compact open sets. This follows from (1) by the Rayleigh characterization. Indeed, enclose the support of a smooth compactly supported test function in finitely many smooth components D_i inside the given open set Ω. Applying (1) on each component gives

$$\int_\Omega |\nabla\varphi|^2 \, d\mu \ge \sum_i c\,\mathrm{Vol}(D_i)^{-2/N} \int_{D_i} |\varphi|^2 \, d\mu \ge c\,\mathrm{Vol}(\Omega)^{-2/N} \int_\Omega |\varphi|^2 \, d\mu.$$

Thus the source assumptions apply also with the smooth-domain convention used here. When $k = 0$, a larger positive Ricci-bound constant meets the source's convention.

We use the following results for bounded integrable data. Set

$$\alpha = \frac{N}{2s + N(m-1)}, \qquad \beta = \frac{2s}{2s + N(m-1)}. \tag{8}$$

Lemma 2.1 (The bounded-data semigroup)**.** *For every nonnegative* $v \in L^1(M) \cap L^\infty(M)$ *there is a mild solution* $S_t v \in C([0,\infty); L^1(M))$ *of* (2). *These solutions form a positivity-preserving semigroup and, for every pair of such data* v, z, *satisfy*

$$\|S_t v - S_t z\|_1 \le \|v - z\|_1, \qquad \|S_t v\|_p \le \|v\|_p \quad (1 \le p \le \infty), \tag{9}$$

and obey

$$\|S_t v\|_\infty \le C_0 \max\{t^{-\alpha}\|v\|_1^\beta, \|v\|_1\}, \qquad t>0, \tag{10}$$

where $C_0 = C_0(N,k,c,s,m)$. Their implicit Euler approximations satisfy

$$z_j + hB(z_j^m) = z_{j-1}, \qquad z_j^m \in D(B \text{ on } L^2(M)), \tag{11}$$

and converge uniformly on each finite time interval in $L^1(M)$.

Proof. The semigroup, contraction, norm estimates and Euler construction are [1, Proposition 5.1 and (5.5)–(5.7)]. In particular, the proof there establishes the stated L^2 operator domain after a positive zeroth-order regularization. Proposition 5.3 identifies these mild solutions as weak dual solutions. The construction in Section 6.1 starts from $\mathbf{1}_{B_j}(v\wedge j)$, where B_j is the geodesic ball of radius j about a fixed point. For bounded compactly supported v this sequence is eventually v. For general bounded integrable v, the contraction identifies the limit with $S_t v$. Consequently [1, Theorem 2.6, (2.13)] gives (10) for this same semigroup. The numbered source statements here refer to the version specified in the bibliography. □

For the energy and time estimates, put

$$V = D(L^{s/2}), \qquad \|v\|_V^2 = \|v\|_2^2 + \left\|L^{s/2}v\right\|_2^2, \qquad \mathcal{E}(v,z) = \langle L^{s/2}v, L^{s/2}z\rangle_2.$$

The form defines $B: V \to V^*$ using the L^2 pivot space. On its operator domain this is the spectral operator already defined above.

Lemma 2.2 (Energy and scalar testing)**.** *Let $u(t) = S_t v$ for nonnegative $v \in L^1\cap L^\infty$. For $0 \le a < b < \infty$,*

$$\int_a^b \mathcal{E}(u^m, u^m)\,dt + \frac{\|u(b)\|_{m+1}^{m+1}}{m+1} \le \frac{\|u(a)\|_{m+1}^{m+1}}{m+1}. \tag{12}$$

In particular $u^m \in L^2((a,b);V)$ and $\partial_t u = -B(u^m)$ in $L^2((a,b);V^*)$. For every $f \in D(B)\cap L^\infty(M)$ such that $Bf \in L^\infty(M)$,

$$\langle u(t) - v, f\rangle = -\int_0^t \langle u(r)^m, Bf\rangle\,dr, \qquad t \ge 0. \tag{13}$$

Proof. The domain assertion in (11) permits an L^2 pairing with z_j^m. Convexity gives

$$\frac{\|z_j\|_{m+1}^{m+1} - \|z_{j-1}\|_{m+1}^{m+1}}{m+1} + h\mathcal{E}(z_j^m, z_j^m) \le 0.$$

For this fixed datum the Euler curves have a common bound $K_v = \|v\|_\infty$. Uniform L^1 convergence and $\int |z-u|^p \le (2K_v)^{p-1}\|z-u\|_1$ give convergence in every finite L^p, uniformly on bounded time intervals. The powers therefore converge strongly in space-time L^2. Their form-energy bound gives a weak limit in $L^2((a,b);V)$, necessarily u^m. Lower semicontinuity proves (12), first from time zero and then after a semigroup restart. The L^2 part of the V norm is bounded by $\|u(t)^m\|_2^2 \le \|u(t)\|_1 \|u(t)\|_\infty^{2m-1}$.

The piecewise affine Euler interpolation has derivative $-B(z_j^m)$. Its difference from the piecewise constant interpolation tends to zero in $L^2((a,b);V^*)$, by integrating that derivative on each step. Testing in time against smooth scalar functions and in space against V identifies the limit derivative as $-B(u^m)$.

For (13), pair the exact Euler equations directly with f. Self-adjointness gives $\langle z_j - z_{j-1}, f\rangle = -h\langle z_j^m, Bf\rangle$. The power map is Lipschitz on $[0,K_v]$, so the right sums converge in L^1 against the bounded function Bf. An incomplete last time step has absolute value at most $h\|Bf\|_\infty K_v^{m-1}\|v\|_1$. The left side converges against bounded f. This proves (13) at every time, by L^1 continuity. □

For later use, if $f \in D(L)$, spectral calculus gives

$$Bf = c_s \int_0^\infty (f - P_r f) \frac{dr}{r^{1+s}}, \qquad c_s = \frac{s}{\Gamma(1-s)}. \tag{14}$$

The integral converges in L^2. If also f, Lf are bounded, then

$$\|f - P_r f\|_\infty \le \min\{2\|f\|_\infty, r\|Lf\|_\infty\}$$

shows that Bf is bounded. In particular every $C_c^\infty(M)$ function is permitted in (13). This also gives the distributional equation (3) for the bounded-data solutions.

Choose nonnegative $v_n \in C_c^\infty(M)$ of integral A, supported in a neighborhood of o of radius tending to zero, all within one compact set K_0. A normalized bump in a fixed coordinate chart gives such a sequence. Let

$$u_n(t) = S_t v_n, \qquad w_n = u_n^m, \qquad M_A(t) = C_0 \max\{A^\beta t^{-\alpha}, A\}.$$

Then $\|u_n(t)\|_1 \le A$ and $\|u_n(t)\|_\infty \le M_A(t)$. Since $1 - \alpha(m-1) = \beta > 0$, we have the uniform estimate

$$\int_0^t \|w_n(r)\|_1 \, dr \le F_A(t) := C_0^{m-1} \left(\frac{A^{1+\beta(m-1)}}{\beta} t^\beta + A^m t \right), \qquad F_A(t) \longrightarrow 0 \quad (t \downarrow 0). \tag{15}$$

For every fixed $0 < a < b$, Lemma 2.2 and smoothing also give

$$\sup_n \left(\|w_n\|_{L^2(a,b;V)} + \|\partial_t u_n\|_{L^2(a,b;V^*)} \right) < \infty. \tag{16}$$

3 Spatial tightness from the linear heat flow

Local convergence on M leaves open the possibility that part of the approximating mass escapes to infinity. We obtain tightness by testing with functions whose fractional Laplacian is uniformly small.

Lemma 3.1. *Let $0 \le \chi_j \le 1$ be smooth compactly supported functions increasing to one. For $H > 0$ define*

$$f_{H,j} = \frac{1}{H} \int_0^H P_r \chi_j \, dr. \tag{17}$$

Then $f_{H,j} \in D(L) \cap C_0(M)$, $0 \le f_{H,j} \le 1$, and

$$\|B f_{H,j}\|_\infty \le C_s H^{-s}, \qquad C_s = c_s \left(\frac{1}{1-s} + \frac{1}{s} \right). \tag{18}$$

For fixed H, the functions $f_{H,j}$ increase to one uniformly on each compact subset of M.

Proof. In the spectral calculus, the average in (17) has multiplier $(1 - e^{-H\lambda})/(H\lambda)$, with value one at zero. This multiplier and its product with λ are bounded. Thus

$$f_{H,j} \in D(L), \qquad L f_{H,j} = \frac{\chi_j - P_H \chi_j}{H}, \qquad \|L f_{H,j}\|_\infty \le H^{-1}.$$

Positivity and contraction give $0 \le f_{H,j}, P_r f_{H,j} \le 1$. The semigroup law gives

$$P_r f_{H,j} - f_{H,j} = \frac{1}{H} \left(\int_H^{H+r} P_\tau \chi_j \, d\tau - \int_0^r P_\tau \chi_j \, d\tau \right),$$

so $\|f_{H,j} - P_r f_{H,j}\|_\infty \le \min\{1, r/H\}$. Splitting (14) at $r = H$ proves (18). In particular these functions belong to the test class of Lemma 2.2.

Continuity follows by integrating the continuous heat flow over $[\eta, H]$ and then letting $\eta \downarrow 0$, with uniform error at most η/H. If $R = d(x, \operatorname{supp}\chi_j)$, (7) gives

$$f_{H,j}(x) \le \frac{C\,\|\chi_j\|_1}{H} \int_0^H r^{-N/2} e^{-R^2/(Cr)}\, dr.$$

For $R \ge 1$, the last integral is at most

$$e^{-R^2/(2CH)} \int_0^H r^{-N/2} e^{-1/(2Cr)}\, dr,$$

which tends to zero as $R \to \infty$. Completeness makes bounded closed neighborhoods of a compact set compact, so $f_{H,j} \in C_0(M)$. Finally $P_r 1 = 1$ and monotone convergence give $f_{H,j} \uparrow 1$. For any $\eta > 0$, the increasing open sets $\{f_{H,j} > 1 - \eta\}$ cover a given compact set; a finite subcover gives the stated uniform convergence. □

Proposition 3.2. *For every $T < \infty$ and $\varepsilon > 0$, there is a compact set $K \subset M$ such that*

$$\sup_n \sup_{0 \le t \le T} \int_{M \setminus K} u_n(t)\, d\mu \le \varepsilon. \tag{19}$$

Proof. Choose H so large that $C_s H^{-s} F_A(T) \le \varepsilon/4$. Then choose j so large that $A \sup_{K_0}(1 - f_{H,j}) \le \varepsilon/4$. Using (13), (15) and (18), for all n and $t \le T$ we get

$$\int_M u_n(t)(1 - f_{H,j})\, d\mu \le A - \int_M u_n(t) f_{H,j}\, d\mu \le \varepsilon/2.$$

There is a compact K outside which $f_{H,j} \le 1/2$. This proves (19). □

4 Compactness and the initial trace

Lemma 4.1. *A subsequence of u_n converges locally in space-time L^{2m} on $(0,\infty)\times M$. There is one set of full measure in positive time on which the same subsequence converges in L^1 on every compact subset of M.*

Proof. Fix $0 < a < b$. From (16), for $0 < h < b - a$,

$$\|u_n(\cdot + h) - u_n\|_{L^2(a, b-h; V^*)} \le Ch.$$

Pair this difference with $w_n(t + h) - w_n(t)$ in V and use Cauchy–Schwarz to get

$$0 \le \int_a^{b-h} \int_M (u_n(t + h) - u_n(t))(w_n(t + h) - w_n(t))\, d\mu\, dt \le Ch. \tag{20}$$

The pairing is the usual integral, since $u_n(t) \in L^2(M)$. On the common range $[0, M_A(a)]$,

$$|p^m - q^m|^2 \le m M_A(a)^{m-1}(p - q)(p^m - q^m).$$

Consequently the powers satisfy the global time-translation bound

$$\int_a^{b-h} \|w_n(t + h) - w_n(t)\|_2^2\, dt \le Ch. \tag{21}$$

For a compact set K and $\delta > 0$, the heat map $P_\delta : L^2(M) \to L^2(K)$ has square-integrable kernel, because

$$\int_K \int_M p_\delta(x,y)^2 \, d\mu(y) \, d\mu(x) = \int_K p_{2\delta}(x,x) \, d\mu(x) < \infty.$$

It can therefore be approximated in operator norm by finite-rank maps: approximate the kernel in product L^2 by finite sums of separated functions and apply Cauchy–Schwarz. On the other hand, the scalar inequality $|1 - e^{-x}|^2 \le C_s x^s$ gives

$$\|v - P_\delta v\|_2^2 \le C_s \delta^s \mathcal{E}(v,v), \qquad v \in V. \tag{22}$$

Thus w_n is uniformly approximated in $L^2((a,b) \times K)$ by the heat-smoothed family and then by a finite-rank family. Its coefficients inherit (21), since they are bounded linear functionals on the global space $L^2(M)$.

For completeness, a coefficient vector $q(t)$ can be replaced by its averages q_I on equal time intervals of length ℓ dividing $b - a$. Its squared approximation error satisfies

$$\sum_I \int_I |q - q_I|^2 = \sum_I \frac{1}{2\ell} \int_I \int_I |q(t) - q(r)|^2 \, dt \, dr \le \frac{1}{\ell} \int_0^\ell \int_a^{b-h} |q(t+h) - q(t)|^2 \, dt \, dh \le C\ell.$$

For a fixed partition these averages form a bounded finite-dimensional family. Choosing successively δ, spatial rank and ℓ proves local space-time L^2 precompactness of w_n.

A diagonal argument over compact spatial sets and positive compact time intervals gives $w_n \to w$ locally in L^2 and almost everywhere. The inequality $|p-q|^m \le |p^m - q^m|$ for $p, q \ge 0$ shows that $u_n \to u = w^{1/m}$ locally in L^{2m}. Choose a further subsequence with summable L^1 errors on every member of a countable space-time exhaustion. Fubini then gives a single full-measure set of times with local spatial L^1 convergence on all compact sets. Intersecting with the full-measure set of time slices of the almost-everywhere space-time convergence preserves this property. □

Proof of Theorem 1.1. At every good time t of Lemma 4.1, local L^1 convergence and Proposition 3.2 imply global L^1 convergence. Indeed, Fatou gives the same small tail for the limit, and the two tails can be made arbitrarily small before using convergence on the remaining compact set. Choose good times $a_j \downarrow 0$. The semigroup property and (9) give

$$\sup_{t \ge a_j} \|u_n(t) - u_\ell(t)\|_1 \le \|u_n(a_j) - u_\ell(a_j)\|_1 \longrightarrow 0.$$

Completeness of $C([a_j, T]; L^1(M))$ therefore gives consistent limits on overlapping intervals. They define one curve $u \in C((0,\infty); L^1(M))$, agreeing with the local limit, and

$$u_n \longrightarrow u \quad \text{uniformly in } L^1(M) \text{ on every } [a,T] \subset (0,\infty). \tag{23}$$

Nonnegativity, $\|u(t)\|_1 \le A$ and $\|u(t)\|_\infty \le M_A(t)$ pass to this representative for every $t > 0$. On $[a,T]$ the power map has a common Lipschitz constant, so (23) also holds globally in L^1 for $u_n^m \to u^m$.

For $\phi \in C_c^\infty((0,\infty) \times M)$, its time support lies in some $[a,T]$. Equation (14) gives

$$\sup_t \|B\phi(t)\|_\infty < \infty.$$

We can therefore pass to the limit in the approximate distributional equations using the global power convergence. This proves (3). The right side is absolutely integrable, since $\|u(t)^m\|_1 \le A M_A(a)^{m-1}$ on $[a,T]$. The same bounds give all the local integrability and boundedness in (4).

For a fixed $\psi \in C_c^\infty(M)$, (13) and (15) imply

$$\left| \int_M u_n(t)\psi \, d\mu - \int_M v_n \psi \, d\mu \right| \le \|B\psi\|_\infty \, F_A(t).$$

Let $n \to \infty$ at fixed $t > 0$, using (23) and the initial concentration at o. Then let $t \downarrow 0$. Since $F_A(t) \to 0$, we obtain $\int_M u(t)\psi \, d\mu \to A\psi(o)$. Uniform approximation of compact continuous functions by smooth compact functions extends this trace to $C_c(M)$, because the mass is bounded by A at every positive time.

Fix $0 \le \chi \le 1$, $\chi \in C_c^\infty(M)$, equal to one near o. The compact-test trace and positivity give

$$\int_M u(t)\chi \, d\mu \longrightarrow A, \qquad 0 \le \int_M u(t)(1-\chi) \, d\mu \le A - \int_M u(t)\chi \, d\mu \longrightarrow 0.$$

For any prescribed $\psi \in C_b(M)$, apply the compact-continuous trace to $\psi\chi$ and bound the remaining integral by $\|\psi\|_\infty \int_M u(t)(1-\chi) \, d\mu$. This proves (6) as an actual limit. The same inequalities give $\|u(t)\|_1 \to A$ as $t \downarrow 0$; together with its upper bound this proves (5) for every $T > 0$. □

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References

[1] E. Berchio, M. Bonforte, G. Grillo and M. Muratori, *The fractional porous medium equation on noncompact Riemannian manifolds*, Mathematische Annalen **389** (2024), 3603–3651. doi:10.1007/s00208-023-02731-6. The numbered source statements used here are in arXiv:2109.10732v2 (2023).

[2] A. Grigor'yan, *Analytic and geometric background of recurrence and non-explosion of the Brownian motion on Riemannian manifolds*, Bulletin of the American Mathematical Society **36** (1999), 135–249. Author's version.

[3] G. Grillo, M. Muratori and F. Punzo, *Fractional porous media equations: existence and uniqueness of weak solutions with measure data*, Calculus of Variations and Partial Differential Equations **54** (2015), 3303–3335. doi:10.1007/s00526-015-0904-4. arXiv:1312.6076v3.

[4] G. Grillo, *Fractional nonlinear diffusions on manifolds: well-posedness and smoothing effects*, contribution to *Mini-Workshop: Hardy Inequalities in Discrete and Continuum Settings*, Oberwolfach Report 14/2025, 597–598. Workshop report.